

Wine Quality Prediction & Analytics Dashboard Project

1. Project Title

Wine Quality Prediction and Analytics Using SQL, Power BI, and Machine Learning

2. Introduction

This project focuses on analyzing wine quality using chemical characteristics and visual analytics. The dataset contains several physicochemical properties such as acidity, density, pH, sulphates, and alcohol content. Using SQL for data analysis and Power BI for dashboard visualization, the project aims to identify the major factors influencing wine quality and generate actionable insights. Additionally, machine learning models can be applied to predict wine quality based on these chemical attributes.

3. Project Objectives

- Analyze wine chemical properties and quality ratings.
- Perform data cleaning and preprocessing.
- Build SQL queries for analytical insights.
- Create interactive Power BI dashboards.
- Identify relationships between alcohol, acidity, and wine quality.
- Generate KPIs and business insights.
- Support data-driven decisions in wine production and quality assessment.

4. Dataset Description

Dataset Source: Kaggle Wine Quality Dataset

The dataset contains physicochemical test results of wine samples along with quality ratings. Each row represents one wine sample.

Main Features:

- Fixed Acidity
- Volatile Acidity
- Citric Acid
- Residual Sugar
- Chlorides
- Free Sulfur Dioxide
- Total Sulfur Dioxide
- Density

- pH
- Sulphates
- Alcohol
- Quality (Target Variable)

5. Tools & Technologies Used

- SQL (MySQL/PostgreSQL) – Data storage and analytical queries
- Power BI – Dashboard development and visualization
- Python – Data preprocessing and machine learning
- Pandas & NumPy – Data manipulation
- Matplotlib & Seaborn – Data visualization
- Scikit-learn – Machine learning algorithms
- DAX – KPI calculations in Power BI

6. Data Cleaning & Preparation

The following preprocessing steps were performed:

- Checked for missing values and duplicates
- Standardized column names
- Verified data types for numerical columns
- Removed inconsistent or invalid records
- Loaded cleaned dataset into SQL database
- Connected SQL database to Power BI
- Created calculated measures using DAX

7. Key KPIs (Using DAX)

1. Total Wines

Total Wines = COUNT(wine_quality[quality])

2. Average Quality

Average Quality = AVERAGE(wine_quality[quality])

3. Average Alcohol

Average Alcohol = AVERAGE(wine_quality[alcohol])

4. Premium Wines

Premium Wines = CALCULATE(COUNT(wine_quality[quality]), wine_quality[quality] >= 7)

5. Premium Wine Percentage

Premium Percentage = DIVIDE([Premium Wines], [Total Wines]) * 100

8. Dashboard Overview

The Power BI dashboard includes:

- KPI cards displaying total wines, average quality, and premium wine percentage
- Bar chart for wine quality distribution
- Line chart showing alcohol vs quality relationship
- Column chart analyzing volatile acidity by quality
- Scatter plots for feature relationships
- Interactive slicers for filtering wine quality and alcohol ranges
- Top wines table sorted by alcohol content

9. Key Insights & Findings

- Higher alcohol content is strongly associated with better wine quality.
- Wines with lower volatile acidity generally receive higher ratings.
- Premium wines form a smaller percentage of the dataset.
- Sulphates and citric acid positively influence wine quality.
- Density tends to decrease slightly in high-quality wines.
- Most wines fall into medium quality categories.

10. Business Impact

This project provides valuable insights for wine manufacturers and quality analysts. By understanding the chemical properties affecting wine quality, producers can optimize production processes and improve product consistency. The dashboard enables quick decision-making through real-time visual analytics and KPI monitoring.

11. Conclusion

The Wine Quality Analytics project demonstrates the practical application of SQL, Power BI, and machine learning in the beverage industry. The project successfully identifies important factors affecting wine quality and presents insights through interactive dashboards and KPIs. The analysis supports data-driven decision-making for wine quality improvement.

12. Future Enhancements

- Implement machine learning prediction models directly in Power BI
- Add real-time data refresh using cloud databases
- Integrate advanced statistical analysis and forecasting
- Develop a web-based dashboard using Streamlit or Flask
- Include customer reviews and sales data for business analytics
- Apply deep learning techniques for improved prediction accuracy

13. Dashboard

